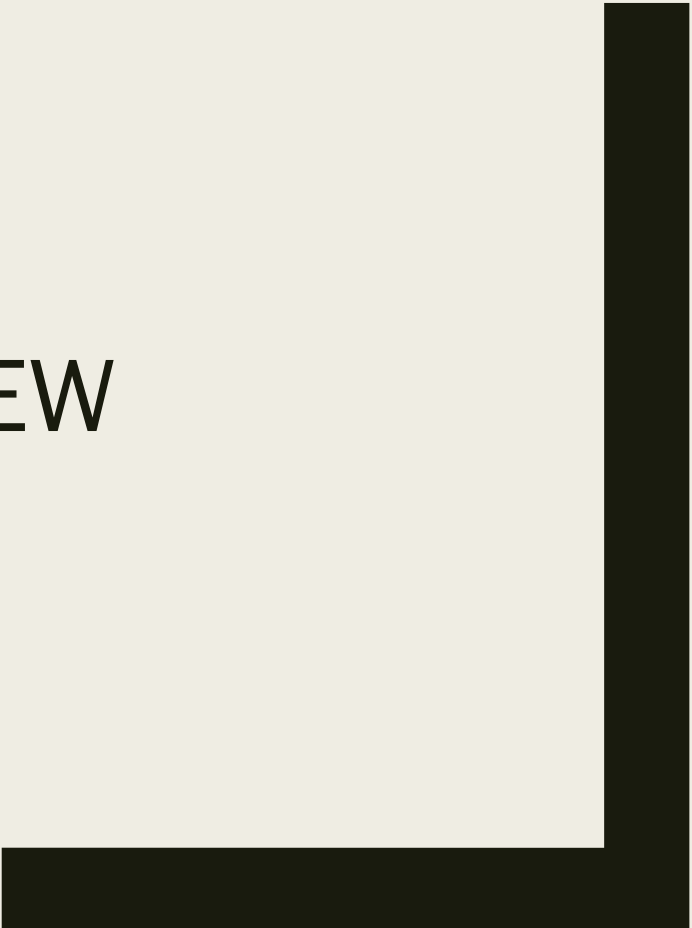




BLOCKCHAIN OVERVIEW



How it all Came About

- It all started with a short paper (8 pages, 8 references)

[1-1-Satoshi Paper - Bitcoin.pdf](#)

- The paper was published by an anonymous person on an anonymous website in 2008
- The title of the paper: “**Bitcoin: A Peer-to-Peer Electronic Cash System**”
- The peer to peer network became operational in 2009
- The technologies used in building the network is not new, but the innovation is in the way they have been put together: distributed protocols, cryptography, a decentralized distributed database (The ledger)

Reading through the Paper:

Motivations & Objectives

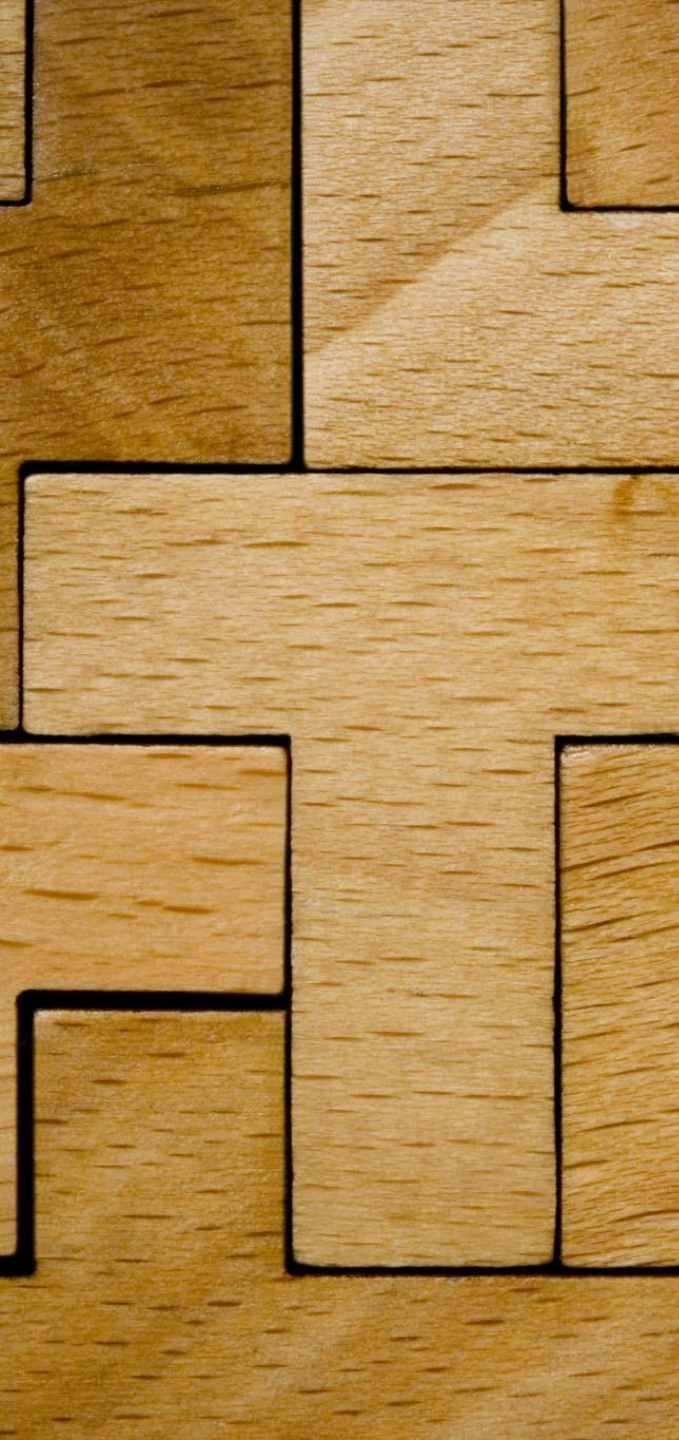
- purely **peer-to-peer** version of electronic cash that makes it possible to send payments directly from one party to another without going through a financial institution.
- Use **digital signatures** to authorize payments
- Prevent **double spending** – without the need for intermediate **trust** entities or organizations.
- A solution to the double spending problem by using a peer-to-peer network
- Timestamp **transactions** by hashing them into an ongoing chain of **hash-based proof-of-work**,
- forming a record that cannot be changed without redoing the **proof-of-work**.

The Peer to Peer Network

- A Peer to Peer (P2P) network is a network where groups of independent computers are interconnected, sharing data without the assistance of any centralized servers.
- Usually deployed on top of the internet.
- All nodes in this type of network are peers with equal responsibility within the network.
- Since there are no special nodes in a P2P network, each peer is both a service provider and a consumer.

Block Structure in Bitcoin

Field	Description	Size
Magic no	The value is always <code>0xD9B4BEF9</code>	4 bytes
Block Size	Number of bytes up to the end of the block	4 bytes
Block Header	Consists of six items	80 bytes
Transaction Counter	Integer count	1 - 9 bytes
Transactions	List of transactions	Variable



Field	Purpose	Size
Version	Block version	4 bytes
hashPrevBlock	256-bit hash of the previous block header	32 bytes
hashMerkleRoot	256-bit hash based on all of the transactions in the block	32 bytes
Time	Current timestamp in seconds since 1970-01-01T00:00 UTC	4 bytes
Difficulty Bits	Current target in a compact format (https://en.bitcoin.it/wiki/Target)	4 bytes
Nonce	32-bit number (starts at 0)	4 bytes

The Hash Fields in the Header

- Hash of the header of the previous block –this is the link between blocks
- Hash of all transactions in the block – if any transaction is modified the hash will change, and the block will be rejected

Peer to Peer Network

- All blockchain systems are deployed on a special type of virtual Peer to Peer network architectures on the Internet

Digital Signatures

- Blockchain systems use public key cryptography to sign transactions and derive account addresses using user public keys

Double Spending

- A common problem in trading and commerce systems
- Traditionally solved using clearances and guarantees provided by financial institutions (takes time and costs a lot)

Transactions

- Blockchain systems are distributed transactional systems
- The transaction is the central entity in blockchain systems

Hash-Based Proof of Work

- Refers to the consensus protocol used in blockchain systems
- The Proof-of-Work consensus protocol is the major innovation in blockchain
- Nodes compete to solve a hard problem (NP Complete) to confirm a block
- All mining nodes compete to compute a special hash of all transactions in a block
- The hash should satisfy a pre-specified format (the difficulty of the problem)

What is Blockchain?

“**Blockchain** at its core is a peer-to-peer distributed ledger that is cryptographically secure, append-only, immutable (extremely hard to change), and updateable only via consensus or agreement among peers.”

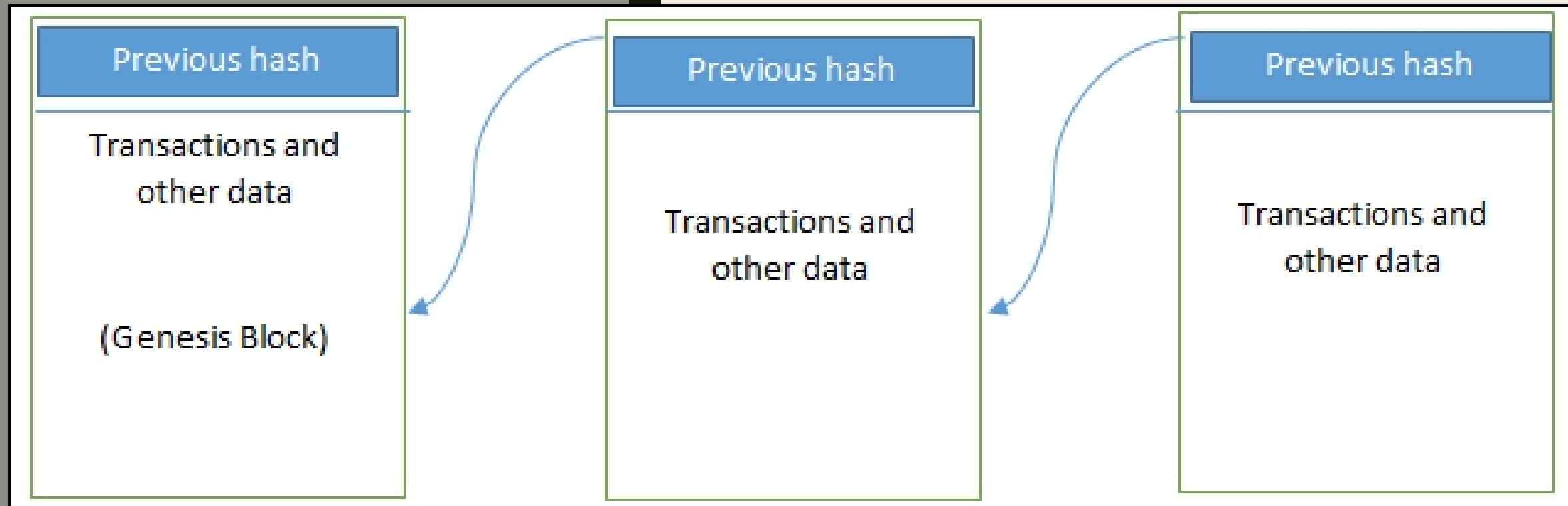
Source: “ Mastering Blockchain”

Many other Definitions

- Sometimes the term “Blockchain” is used to refer to the underlying distributed secure “peer-to-peer” network
- It is also used to refer to the consensus mechanism
- However, the rising trend now is that the term Blockchain refers to the whole technology associated with cryptocurrency systems, including algorithms, data structures, underlying network ...etc
- Accordingly, the term “Blockchain Technology Systems” may be more appropriate

The Blockchain Structure

- Linked Blocks



Some Example Transactions from Bitcoin

Transaction 7957a35fe64f80d234d76d83a2a8f1a0d8149a41d81de548f0a65a8a999f6f18

INPUTS From		OUTPUTS To	
From (previous transactions Joe has received):		Output #0 Alice's Address	0.1000 BTC (u)
Joe	0.1005 BTC	Transaction Fees:	0.0005 BTC

Transaction 0627052b6f28912f2703066a912ea577f2ce4da4caa5a5fbd8a57286c345c2f2

INPUTS From		OUTPUTS To	
7957a35fe64f80d234d76d83a2a8f1a0d8149a41d81de548f0a65a8a999f6f18 : 0		Output #0 Bob's Address	0.0150 BTC (u)
Alice	0.1000 BTC	Output #1 Alice's Address (change)	0.0845 BTC (u)
		Transaction Fees:	0.0005 BTC

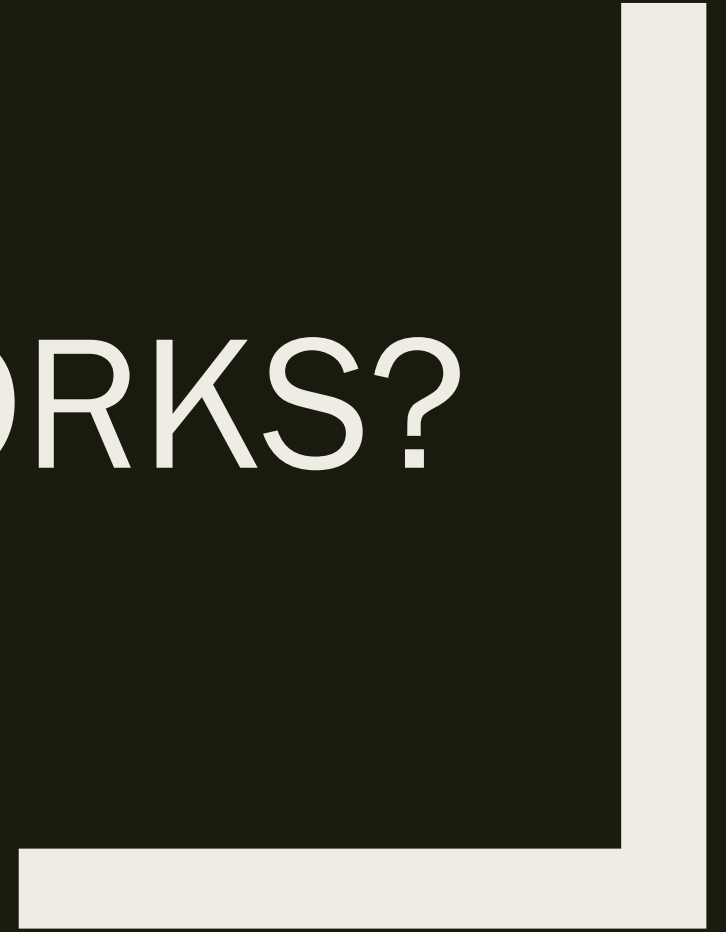
Transaction 2bbac8bb3a57a2363407ac8c16a67015ed2e88a4388af58cf90299e0744d3de4

INPUTS From		OUTPUTS To	
0627052b6f28912f2703066a912ea577f2ce4da4caa5a5fbd8a57286c345c2f2 : 0		Output #0 Gopesh's Address	0.0100 BTC (u)
Bob	0.0150 BTC	Output #1 Bob's Address (change)	0.0045 BTC (u)
		Transaction Fees:	0.0005 BTC

Block Structure

Size	Field	Description
4 bytes	Block Size	The size of the block, in bytes, following this field
80 bytes	Block Header	Several fields form the block header
1–9 bytes (VarInt)	Transaction Counter	How many transactions follow
Variable	Transactions	The transactions recorded in this block

HOW IT WORKS?



Transaction Life Cycle

- A user/sender generates a transaction using a wallet software or some other interface.
- The wallet software signs the transaction using the sender's private key.
- The transaction is broadcasted to the blockchain network using a flooding algorithm.
- Mining nodes include this transaction in the next (candidate) block to be mined.
- Mining starts on a new block once the last (previous) has been broadcasted by a (winner) node that succeeded in solving the “Proof of Work.”

Transaction Life Cycle

- The nodes verify the transaction and propagate the transaction further, and confirmation starts to generate.
- Each mining node creates a block and add transactions to the block
- Miner nodes compete to verify the block
- The first node that confirms the block (through proof of work) broadcasts the confirmed block
- All other nodes stop mining and focus on verifying the confirmed block
- All transactions in the block are added to the ledger

The Mining Process

- The Mining process is performed approximately, every 10 minutes in Bitcoin
- All transactions generated during this period are packed into a “Block”, simultaneously on every mining node. (every node generates its version of the block – only the block of the winner node is propagated)
- Miner nodes compete to mine that Block (every node tries to generate a cryptographic hash of the block header that matches a “target” solution pre-specified by the system.
- This is usually a difficult (hard) problem that requires considerable time and computational resources. Solving this problem is the “Proof of Work (POW)”, which strengthens the security of the network.

The Mining Process

- Although thousands of nodes will be working in parallel (every 10 minutes, on average) to mine the current block, eventually there is only one winner: the first node to propagate the mined block.
- On receiving the winning block, all other nodes stop working on the current block, and immediately start working on the next block.

Emergent Consensus

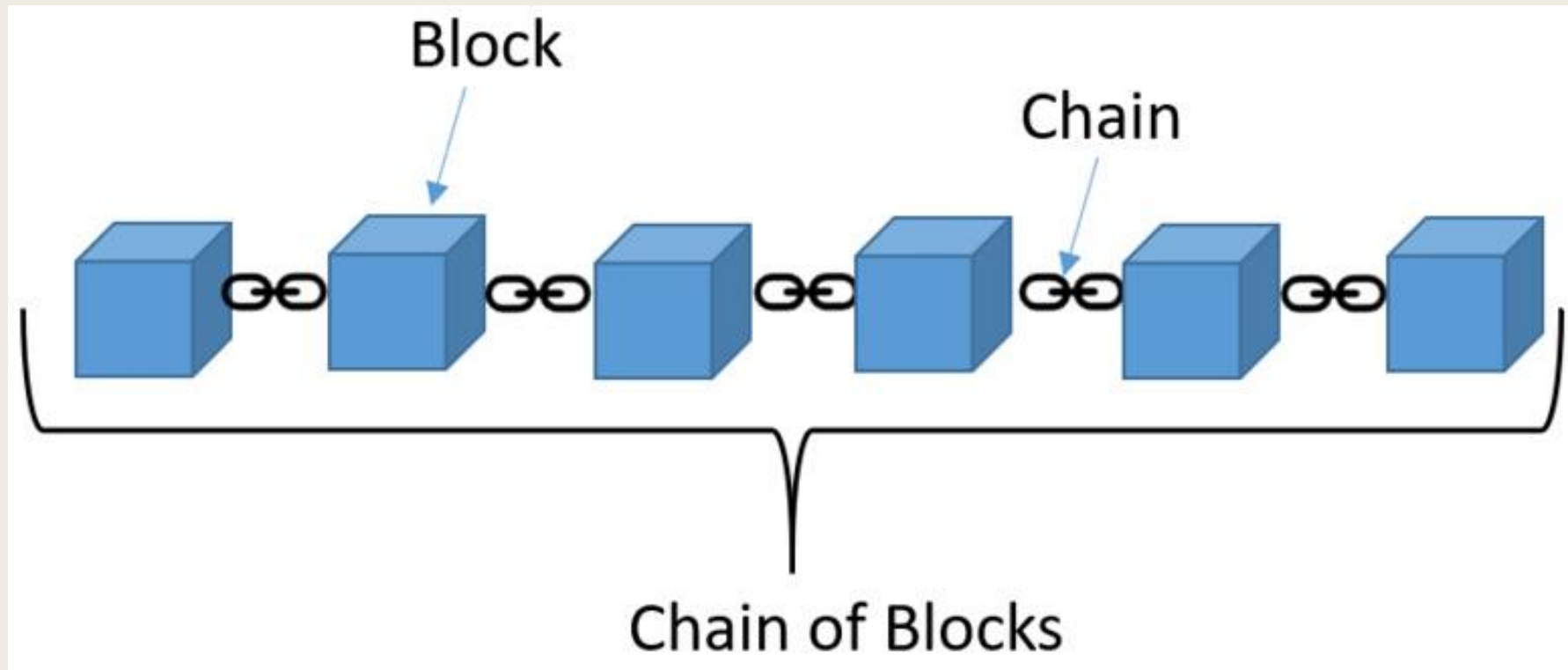
Consensus emerges as a result of four processes:

- Independent verification of each transaction, by every full node, based on a list of criteria
- Independent aggregation of those transactions into new blocks by mining nodes, coupled with demonstrated computation through a Proof-of-Work algorithm
- Independent verification of the new blocks by every node and assembly into a chain
- Acceptance by every node, of the chain with the most cumulative computation demonstrated through Proof-of-Work

Features

- **Decentralization:** In simple terms, this means that the application or service continues to be available and usable even if a server or a group of servers on a network crashes or is not available. The service or application is deployed on a network in a way that no server has absolute control over data and execution; rather, each server has a current copy of data and execution logic.
- **Distributed:** This means that any server or node on a network is connected to every other node on the network. Rather than having one-to-one or one-to-many connectivity between servers, servers have many-to-many connections with other servers.
- **Database:** This refers to the location for storing durable data that can be accessed at any point in time. A database allows the storage and retrieval of data as functionality and also provides management functionalities to manage data efficiently, such as exporting, importing, backup, and restoration.

Architectutre



Architecture

